generalized Eigenspace $\{ v \in \mathbb{R}^n \mid (\lambda I - A)^k v = 0 \text{ for some } k \} = \widetilde{E_\lambda}$.

Claim: $\sum_\lambda \dim(\widetilde{E_\lambda}) = n$. $\widetilde{E_\lambda}$.

MA1522 Linear Algebra for Computing

Lesson 17: Diagonalization and Orthogonal Diagonalization

Cayley–Hamilton thm: $ch_A(x) = (x-\lambda_1)^{r_1} \cdots (x-\lambda_s)^{r_s}$.

$(A-\lambda_1 I)^{r_1} \cdots (A-\lambda_s I)^{r_s} = 0$

$(A-\lambda_1 I)^{r_1} \cdots (A-\lambda_s I) \cdot v = 0 \quad \forall v$.

with some work $\Longrightarrow$ claim..

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson17` (e.g. `A1234567Z_lesson17.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 17**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Example 1

True or false? Let A be a square matrix and $\lambda_1 \neq \lambda_2$ be two distinct eigenvalues of A. Let S_1 and S_2 be bases of the eigenspaces E_{λ_1} and E_{λ_2} respectively. Then

1. $S_1 \cup S_2$ is linearly independent. True $\Longleftrightarrow E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.
2. S_1 is orthogonal to S_2. False, unless A is symmetric.

1. Suppose $a_1 u_1 + a_2 u_2 + \cdots + a_k u_k + (b_1 v_1 + \cdots + b_l v_l) = 0$

$\Longleftrightarrow \underbrace{a_1 u_1 + \cdots + a_k u_k}_{\in E_{\lambda_1}} = \underbrace{-b_1 v_1 - \cdots - b_l v_l}_{\in E_{\lambda_2}} = w \in E_{\lambda_1} \cap E_{\lambda_2}$.

$w \in E_{\lambda_1}$, $w \in E_{\lambda_2}$

$\lambda_1 w = Aw = \lambda_2 w$

$\Downarrow$

$(\lambda_1 - \lambda_2) w = 0 \Rightarrow w = 0$ ($\lambda_1 - \lambda_2 \neq 0$)

$\Rightarrow a_1 = \cdots = a_k = 0$, $b_1 = \cdots = b_l = 0$

S_1 and S_2 are linearly independent.

2. $A = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$. $E_0 = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}$. $E_1 = \operatorname{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$. $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \neq 0$

Section 6.2 Diagonalization

Let A be a square matrix of order n.

conjugate A by P.

Definition

A is **diagonalizable** if there exists an invertible matrix P such that $P^{-1}AP = D$ is a diagonal matrix, or equivalently, $A = PDP^{-1}$.

Theorem

A is diagonalizable if and only if A has n linearly independent eigenvectors. In that case

$P_{T\to S}$ $S=\{e_1,\dots,e_n\}$, $T=\{u_1,\dots,u_n\}$ $= P = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{pmatrix}, \qquad D = \operatorname{diag}(\mu_1, \mu_2, \dots, \mu_n),$

where $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$ of eigenvectors of A and $A\mathbf{u}_i = \mu_i \mathbf{u}_i$.

$A \cdot u_i = \mu_i u_i$

$P^{-1}AP\, e_i = P^{-1}A\,(u_1 \cdots u_n)\begin{pmatrix}0\\ \vdots \\ 1 \\ \vdots \\ 0\end{pmatrix} = P^{-1}A\,u_i = P^{-1}\mu_i u_i = \mu_i (P^{-1}u_i) = \mu_i e_i$

$\begin{pmatrix}0\\ \vdots \\ 1 \\ \vdots \\ 0\end{pmatrix}$ i-th position

u_i

Conclusion since $P^{-1}APe_i = \mu_i e_i$.

$P^{-1}AP = \begin{pmatrix}\mu_1 & & \\ & \ddots & \\ & & \mu_n\end{pmatrix}$

Section 6.2 Equivalent Statements for Diagonalizability

char poly of $A = (x-\lambda_1)^{r_{\lambda_1}} \cdots (x-\lambda_s)^{r_{\lambda_s}}$.

deg n.

$\Rightarrow \quad \sum_{\lambda} r_\lambda = n.$

Theorem

For every eigenvalue λ of A, $\quad 1 \le \dim(E_\lambda) \le r_\lambda$.

Theorem

A is diagonalizable if and only if

1. the characteristic polynomial splits into linear factors, over $\mathbb{R}$. $\det(xI - A) = (x-\lambda_1)^{r_{\lambda_1}} \cdots (x-\lambda_k)^{r_{\lambda_k}}$, and λ_i are real numbers
2. $\dim(E_{\lambda_i}) = r_{\lambda_i}$ for every eigenvalue λ_i.

Example: x^2+1 does not split / $\mathbb{R}$.

Section 6.2 Algorithm for Diagonalization

Let A be a square matrix of order n.

$\lambda_i \in \mathbb{R}$.

1. Compute and factorise $\det(xI - A) = (x - \lambda_1)^{r_{\lambda_1}} \cdots (x - \lambda_k)^{r_{\lambda_k}}$. If it does not split into linear factors, stop.

2. For each eigenvalue λ_i, find a basis S_{λ_i} for E_{λ_i}, the solution space of $(\lambda_i I - A)\mathbf{x} = \mathbf{0}$. If $\dim(E_{\lambda_i}) < r_{\lambda_i}$, stop.

3. $S = S_{\lambda_1} \cup \cdots \cup S_{\lambda_k} = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$ of eigenvectors of A.

4. Let $P = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $D = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, where $A\mathbf{u}_i = \mu_i \mathbf{u}_i$. Then $A = PDP^{-1}$.

$P^{-1}AP = D$.

Question 1

Let

$$S = \begin{pmatrix} -4 & 1 & -2 \\ 1 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix}, \qquad \det(xI - S) = (x+5)^2(x-1).$$

1. For each eigenvalue λ of S, find a basis for E_λ.

2. Is S diagonalizable? If so, find an invertible matrix P and a diagonal matrix D such that $S = PDP^{-1}$. Check your answer.

Question 1 — solution: part 1

$S = \begin{pmatrix} -4 & 1 & -2 \\ 1 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix}$ $\quad ch_S(x) = (x+5)^2(x-1)$.

Step 1 char poly splits ✓. $\lambda_1 = -5$, $\lambda_2 = 1$

$r_{\lambda_1} = 2$ $\quad r_{\lambda_2} = 1$.

Step 2 Compute basis for eigenspaces E_{λ_1} $\quad E_{\lambda_2}$

$\| \quad \| $

$\text{Null}(\lambda_1 I - S)$ $\quad \text{Null}(\lambda_2 I - S)$

- $\text{Null}(-5I - S)$

$$-5I - S = \begin{pmatrix} -1 & -1 & 2 \\ -1 & -1 & 2 \\ 2 & 2 & -4 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \left\{ \begin{pmatrix} -t-s \\ t \\ s \end{pmatrix} \,\middle|\, s,t \in \mathbb{R} \right\} = \text{span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\}$$

basis for E_{-5}.

- $\text{Null}(1I - S)$

$$I - S = \begin{pmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \left\{ \begin{pmatrix} -t \\ -t \\ 2t \end{pmatrix} \,\middle|\, t \in \mathbb{R} \right\} = \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right\}$$

basis for E_1.

$2 = \dim(E_{-5}) = r_{-5}$

$1 = \dim(E_1) = r_1$ $\Rightarrow$ S is diagonalizable

Rmk: If you know $\dim(E_{-5}) = 2$, then S is diagonalizable

$1 \le \dim(E_1) \le r_1 = 1$

Question 1 — solution: part 2

2. Is S diagonalizable? If so, find P invertible and D diagonal with $S = PDP^{-1}$. Check your answer.

u_1 u_2

$E_{-5} = \text{span}\left\{\begin{pmatrix}-1\\1\\0\end{pmatrix}, \begin{pmatrix}2\\0\\1\end{pmatrix}\right\}$.

$E_1 = \text{span}\left\{\begin{pmatrix}1\\1\\-2\end{pmatrix}\right\}$.

u_3.

$P = (u_1 \; u_2 \; u_3) = \begin{pmatrix}-1 & 2 & 1\\ 1 & 0 & 1\\ 0 & 1 & -2\end{pmatrix}$.

$D = \begin{pmatrix}-5 & & \\ & -5 & \\ & & 1\end{pmatrix}$

Check $S = PDP^{-1}$

$\Leftrightarrow \quad SP = PD$.

check this.

$$\begin{pmatrix}-4 & 1 & -2\\ 1 & -4 & -2\\ -2 & -2 & -1\end{pmatrix}\begin{pmatrix}-1 & 2 & 1\\ 1 & 0 & 1\\ 0 & 1 & -2\end{pmatrix} = \begin{pmatrix}-1 & 2 & 1\\ 1 & 0 & 1\\ 0 & 1 & -2\end{pmatrix}\begin{pmatrix}-5 & & \\ & -5 & \\ & & 1\end{pmatrix}$$

Question 2

Let a be a real number, and

$$B_a = \begin{pmatrix} 2 & 1 & a \\ 0 & -2 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

1. Find all values of a for which B_a is diagonalizable.

2. For each such a, find an invertible matrix P and a diagonal matrix D such that $B_a = PDP^{-1}$.

Question 2 — solution: part 1

$$B_a = \begin{pmatrix} 2 & 1 & a \\ 0 & -2 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

Step 1: char poly splits ?

$\det \begin{pmatrix} X-2 & -1 & -a \\ 0 & X+2 & -4 \\ 0 & 0 & X-2 \end{pmatrix}$ = det triangular matrix $= (X-2)^2(X+2)$ splits ✓.

Step 2: Compute basis for eigenspace for $\lambda_1 = 2$, $\lambda_2 = -2$.

$\dim$ $r_{\lambda_1} = 2$ $r_{\lambda_2} = 1$.

So

$$1 \leq \dim(E_{\lambda_1}) \overset{?}{\leq} r_{\lambda_1} = 2$$

$$1 \leq \dim(E_{\lambda_1}) \leq r_{\lambda_2} = 1.$$

$\dim(E_{\lambda_1}) = 1$

B_a is diagonalizable $\iff \dim(E_2) = 2$.

E_2 = Null $(2I - B_a)$.

$$2I - B_a = \begin{pmatrix} 0 & -1 & -a \\ 0 & 4 & -4 \\ 0 & 0 & 0 \end{pmatrix}$$

$R_2 + 4R_1 \rightarrow \begin{pmatrix} 0 & -1 & -a \\ 0 & 0 & 4-4a \\ 0 & 0 & 0 \end{pmatrix}$

$\dim(E_2) = 2 \iff$ Multiplicity $(2I - B_a) = 2$

$\iff$ rank $(2I - B_a) = 1 = \dim(\text{Row}(2I - B_a))$

$\iff (0 \ -1 \ -a) \in \text{Span}\{(0 \ 4 \ -4)\}$

$\iff -a = 1$, $a = -1$.

Conclusion: B_a is diagonalizable $\iff a = -1$.

Question 2 — solution: part 2

When $a=-1$, diagonalize it. $\lambda_1 = 2, \quad \lambda_2 = -2$

$$B_{-1} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & -2 & 4 \\ 0 & 0 & 2 \end{pmatrix}.$$

$$2I - B_{-1} = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 4 & -4 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{ref}} \begin{pmatrix} 0 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \text{span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$E_2 = \text{span}\{u_1, u_2\}$

$$-2I - B_1 = \begin{pmatrix} -4 & -1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & -4 \end{pmatrix} \Longrightarrow E_{-2} = \text{span}\left\{ \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix} \right\}$$

u_3

$$P = \begin{pmatrix} u_1 & u_2 & u_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -4 \\ 0 & 1 & 0 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & & \\ & 2 & \\ & & -2 \end{pmatrix}.$$

Homework

For each matrix below, determine whether it is diagonalizable. If it is, find an invertible matrix P and a diagonal matrix D such that it equals PDP^{-1}.

$$H_1 = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix}, \qquad H_2 = \begin{pmatrix} 3 & 0 & 0 \\ 1 & -2 & -3 \\ 4 & 0 & 1 \end{pmatrix}, \qquad H_3 = \begin{pmatrix} -1 & -4 & 1 \\ 0 & 3 & -1 \\ 0 & 1 & 1 \end{pmatrix},$$

$$H_4 = \begin{pmatrix} -1 & 3 & -3 & -3 \\ 0 & -1 & -3 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -3 & -3 & 2 \end{pmatrix}, \qquad H_5 = \begin{pmatrix} 2 & -1 & -3 & 2 \\ 0 & 3 & 0 & 0 \\ 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 \end{pmatrix}.$$

The eigenvalues of $H_2, \ldots, H_5$ are integers.

Section 6.3 Orthogonally Diagonalizable

PDP^{-1}

Definition

A square matrix A of order n is **orthogonally diagonalizable** if $A = PDP^T$ for some orthogonal matrix P (that is, $P^T = P^{-1}$) and diagonal matrix D.

Theorem (Spectral theorem)

A is orthogonally diagonalizable if and only if A is symmetric. Equivalently, there is an orthonormal basis of $\mathbb{R}^n$ of eigenvectors of A.

Theorem

If A is symmetric, $\lambda_1 \neq \lambda_2$ are eigenvalues of A and $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$, then $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$: the eigenspaces of a symmetric matrix are orthogonal to each other.

$v_1 \cdot v_2 = 0$

$\Uparrow$

$(\lambda_1 - \lambda_2) \cdot (v_1 \cdot v_2) = 0$

$\neq 0$

$v_1 \cdot (Av_2) = v_1 \cdot \lambda_2 v_2 = \lambda_2 (v_1 \cdot v_2)$

$(v_1^t A) v_2 = (A^t v_1)^t v_2 = (A^t v_1) \cdot v_2 = (Av_1) \cdot v_2 = \lambda_1 (v_1 \cdot v_2)$

(A is sym)

Section 6.3 Algorithm for Orthogonal Diagonalization

Let A be a symmetric matrix of order n.

1. Compute $\det(xI - A) = (x - \lambda_1)^{r_{\lambda_1}} \cdots (x - \lambda_k)^{r_{\lambda_k}}$.

2. For each eigenvalue λ_i, find a basis S_{λ_i} for E_{λ_i}.

3. Apply the Gram–Schmidt process to each S_{λ_i} to get an orthonormal basis T_{λ_i} of E_{λ_i}. Then $T = T_{\lambda_1} \cup \cdots \cup T_{\lambda_k} = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is an orthonormal basis for $\mathbb{R}^n$.

4. Let $P = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $D = \operatorname{diag}(\mu_1, \ldots, \mu_n)$, where $A\mathbf{u}_i = \mu_i \mathbf{u}_i$. Then P is orthogonal and $A = PDP^T$.

Question 3

Let S be the matrix in Question 1,

$$S = \begin{pmatrix} -4 & 1 & -2 \\ 1 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix}.$$

Orthogonally diagonalize S: find an orthogonal matrix P and a diagonal matrix D such that $S = PDP^T$.

Question 3 — solution

$S = \begin{pmatrix} -4 & 1 & -2 \\ 1 & -4 & -2 \\ -2 & -2 & -1 \end{pmatrix}$ sym. $E_{-5} = \operatorname{span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}$. $E_1 = \operatorname{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \right\}$.

v_1 v_2 v_3

Apply Gram-Schmidt process to eigenspace. to get orthonormal basis

- $v_1 \cdot v_1 = 2$. $u_2 = v_2 - \frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{-2}{2} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ $u_2 \cdot u_2 = 3$

  $u_1 = v_1$

  To make it orthonormal, we get $\left\{ w_1 = \frac{u_1}{\|u_1\|} = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \quad w_2 = \frac{u_2}{\|u_2\|} = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \right\}$.

  orthonormal basis for E_{-5}.

- For E_1 $\left\{ w_3 = \frac{v_3}{\|v_3\|} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \\ \frac{-2}{\sqrt{6}} \end{pmatrix} \right\}$.

- So $P = \begin{pmatrix} w_1 & w_2 & w_3 \end{pmatrix}$ is orthogonal, and $P^{-1}AP = D = \begin{pmatrix} -5 & & \\ & -5 & \\ & & 1 \end{pmatrix}$

Next lesson

Next lesson (**Tue 27 Oct**, Sections 6.4 & 6.5), I will focus on

1. powers of diagonalizable matrices, and Markov chains;

2. the **singular value decomposition**.

Please watch the video clips beforehand.